Natural Computing SoSe24

Exercise Sheet 4 — June 24th, 2025

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1 Ising Model

Let

$$H'(\vec{\sigma}) = -7\sigma_1 + 5\sigma_2 - 15\sigma_3 - 3\sigma_4 + 12\sigma_1\sigma_4 + 6\sigma_1\sigma_2 + 8\sigma_2\sigma_3 + 2\sigma_1\sigma_3$$

be an energy function to be minimized, where $\vec{\sigma} \in \{-1, 1\}^4$. Recall that in the lecture we have covered the function to compute the energy $H(\vec{\sigma})$ of a spin vector $\vec{\sigma}$ given interaction matrix J and field vector $\vec{h}$ in an equivalent way to the following definition:

$$H(\vec{\sigma}) = \sum_{i=1}^{N} \sum_{j>i}^{N} \sigma_i \sigma_j J_{ij} + \sum_{i=1}^{N} h_i \sigma_i$$

(i) Give J and h to define the Ising model for $H' = H$.

(ii) Compute the energies $H(\vec{\alpha}), H(\vec{\beta})$ given J, h, and the following solution candidates

(a) $\vec{\alpha} = (1, 1, 1, 1)$

(b) $\vec{\beta} = (-1, 1, -1, 1)$

(iii) Are the energies for either of the two solution candidates $\vec{\alpha}, \vec{\beta}$ already optimal? If not, find

$$\min_{\vec{x} \in \{-1,1\}^4} H'(\vec{x}),$$

where H' is the energy function from task (i).

2 Ising to QUBO

Let again

$$H'(\vec{\sigma}) = -7\sigma_1 + 5\sigma_2 - 15\sigma_3 - 3\sigma_4 + 12\sigma_1\sigma_4 + 6\sigma_1\sigma_2 + 8\sigma_2\sigma_3 + 2\sigma_1\sigma_3$$

be an energy function to be minimized, where $\vec{\sigma} \in \{-1, +1\}^4$ is our solution candidate. To compute the energy $H(\vec{\sigma})$ of a solution vector $\vec{\sigma}$, given interaction matrix J and field vector $\vec{h}$, we recall the following definition:

$$H(\vec{\sigma}) = \sum_{i=1}^{N} \sum_{j>i}^{N} \sigma_i \sigma_j J_{ij} + \sum_{i=1}^{N} h_i \sigma_i$$

Convert the optimal Ising solution $[1, -1, 1, -1]$ to a QUBO solution and explain why we need the conversion $\sigma_i = 2x_i - 1$ for all i, if we want to keep the energy values consistent.